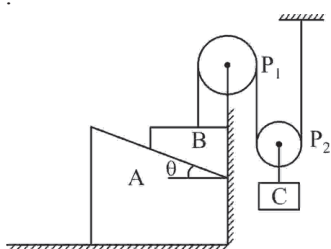


Advanced Physics Assignment-1A| 2019

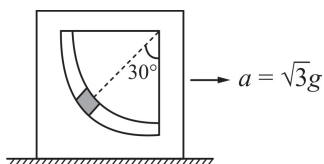
Mechanics

1. In the figure shown P_1 and P_2 are massless pulleys. P_1 is fixed and P_2 can move. Masses of A, B and C are $\frac{9m}{64}$, $2m$ and m respectively. All contacts are smooth and the string is massless, $\theta = \tan^{-1}\left(\frac{3}{4}\right)$. Find the acceleration of block C is m/s^2 .



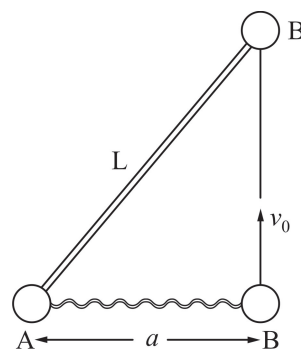
- (A) $2 m/s^2$ (B) $3 m/s^2$
(C) $4 m/s^2$ (D) $5 m/s^2$

2. A wedge with a rough groove in the shape of a quarter of a circle is kept on a smooth table (see figure). A small block is placed in the groove with a small clearance. The wedge is moved with an acceleration $\sqrt{3}g$. If the block is to remain stationary relative to groove in the position shown, the coefficient of friction required can be :



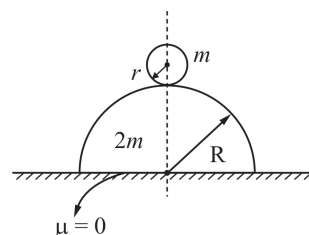
- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$
(C) $\frac{1}{5}$ (D) $\frac{9}{10}$

3. Two identical spheres A and B, each of mass m are attached to an inextensible inelastic cord of length L , and are resting at a distance a from each other on a frictionless horizontal surface. Sphere B is given a velocity v_0 in a direction perpendicular to line AB and moves without until it reaches B' when the cord becomes taut. The energy lost as the cord becomes taut is given by :



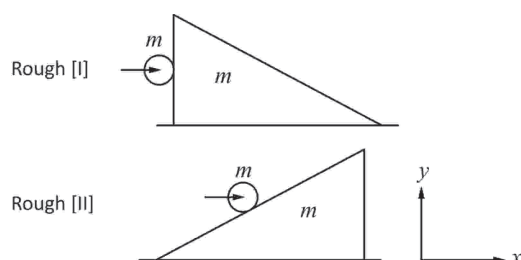
- (A) $\frac{mv_0^2}{2L^2}(L^2 - a^2)$ (B) $\frac{mv_0^2}{4L^2}(L^2 - a^2)$
(C) $\frac{mv_0^2}{L^2}(L^2 - a^2)$ (D) None of these

4. A small sphere of radius r and mass m is placed on a big hemisphere of mass $2m$ and radius R which is kept on a smooth horizontal surface as shown in the figure. Find the displacement of hemisphere when line joining between centre of small sphere and centre of big hemisphere makes 30° with the vertical (initially this line was vertical, and assume small sphere is not losing the contact with hemisphere)



- (A) $\frac{R+r}{2}$ (B) $\frac{R+r}{3}$
 (C) $\frac{R+r}{6}$ (D) None of these

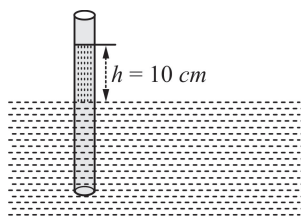
5. A ball of mass m collides horizontally with a stationary wedge on a rough horizontal surface, in the two orientations as shown. Neglect friction between ball and wedge. Two student comment on system of ball and wedge in these situations :



Saurav : Momentum of system in x -direction will change by significant amount in both cases.

Rahul : There are no impulsive external forces in y -direction in both cases hence the total momentum of system in y -direction can be treated as conserved in both cases.

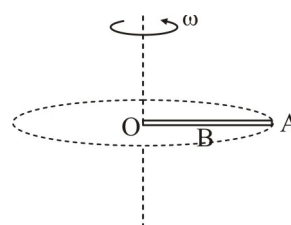
- (A) Saurav is wrong and Rahul is correct
 (B) Saurav is correct and Rahul is wrong
 (C) Both are correct
 (D) Both are wrong
6. A capillary tube is dipped in water vertically. It is sufficient long so that water rises to maximum height 10 cm in the tube. The length of the portion immersed in water is 25 cm. The lower end of the tube is closed, the tube is taken out and opened again. Then :



- (A) All water flows out of the tube
 (B) The length of the water column remaining in the tube will be 10 cm

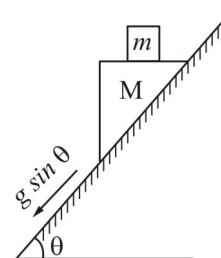
- (C) The length of the water remaining in the tube will be 20 cm
 (D) The length of the water column remaining in the tube will be 35 cm

7. A rod OA of mass m and length L is rotating in a horizontal circle about point O as shown in figure. B is the mid-point of rod. If T represents tension in the rod at any point and U represents elastic potential energy per unit volume at any point of rod, then choose correct alternative(s).



- (A) $T_0 = \frac{4}{3}T_B$ (B) $T_0 = 2T_B$
 (C) $U_0 = \frac{16}{9}U_B$ (D) $U_0 = 4T_B$

8. A wedge of mass M and a block of mass m are kept on an incline of inclination θ . The system is released from rest. The surface of the wedge on which the block rests is horizontal. All the surfaces are smooth. Then :



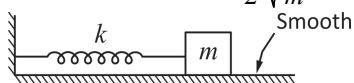
- (A) The acceleration of the block is along the incline
 (B) The acceleration of the block is in vertical direction
 (C) The acceleration of the wedge is greater than $g \sin \theta$
 (D) Normal force exerted by incline plane on wedge is less than $(M + m) g \cos \theta$

9. Figure shows an ideal spring block system, force constant of spring is k which has been compressed by an amount x_0 . If x is instantaneous deflection of spring from its natural length, mark the correct options.

(A) Instantaneous power developed by spring

$$\text{is } P = kx \sqrt{\frac{k}{2m}(x_0^2 - x^2)}$$

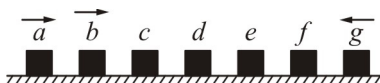
(B) Maximum power of spring is $\frac{k}{2} \sqrt{\frac{k}{m}} x_0^2$



(C) Maximum power of spring is $x = \frac{x_0}{\sqrt{2}}$

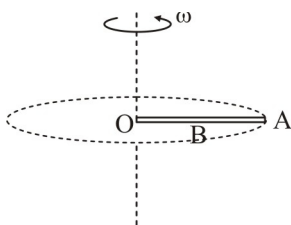
(D) Maximum power occurs at $x = \frac{x_0}{2}$

10. Figure shows seven identical blocks kept equidistant on a frictionless floor. Initially blocks a and b are moving rightward and block g is moving leftward, each with speed $v = 3 \text{ m/s}$. The other blocks are stationary. A series of elastic collisions occurs. After the last collision the speed and direction of motion of:



- (A) A will be at rest
(B) D will be 1.5 m/s towards left
(C) E will be at rest
(D) G will be 3 m/s towards right

11. A rod OA of mass m and length L is rotating in a horizontal circle about point O as shown in figure. B is the mid-point of rod. If T represents tension in the rod at any point and U represents elastic potential energy per unit volume at any point of rod, then choose correct alternative(s).



- (A) $T_0 = \frac{4}{3} T_B$ (B) $T_0 = 2 T_B$
(C) $U_0 = \frac{16}{9} U_B$ (D) $U_0 = 4 T_B$

Paragraph for Questions 12 - 14

A particle starts moving at $t = 0$ in x - y plane such that its coordinates (mm) change with time (in sec) as $x = 2t$ and $y = 10 \sin(2t)$.

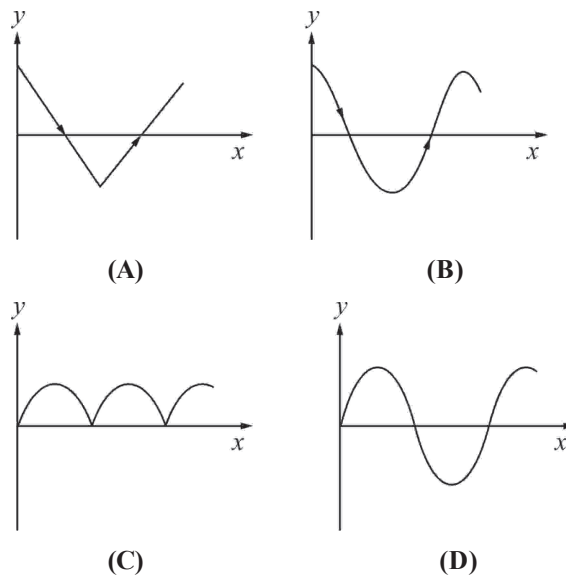
12. If magnitude of its acceleration a , then :

- (A) $a \propto x$ (B) $a \propto y$
(C) $a \propto \sqrt{x^2 + y^2}$ (D) $a = 0$

13. Maximum speed of the particle is :

- (A) 12 m/s (B) $2\sqrt{26} \text{ m/s}$
(C) $2\sqrt{101} \text{ m/s}$ (D) 10 m/s

14. The path of the particle will be :



Paragraph for Questions 15 - 17

A block is tied within two springs, each having spring constant equal to k . Initially the springs are in their natural length and horizontal as shown in the figure, the block is released from rest. The springs are ideal, acceleration due to gravity is g downwards. Air resistance is to be neglect. The natural length of spring is l_0 .



15. If the decrease in height of the block till it reaches equilibrium is $\sqrt{3}I_0$ then the mass of the block is :

(A) $\frac{2kI_0}{g}$ (B) $\frac{\sqrt{2}kI_0}{g}$
 (C) $\frac{\sqrt{3}kI_0}{g}$ (D) None of these

16. When the block is under equilibrium as per the conditions of previous question, then the speed of the block is :

(A) Zero (B) $\sqrt{\frac{4gI_0}{\sqrt{3}}}$
 (C) $\frac{\sqrt{3}kI_0}{g}$ (D) None of these

18. A small ball is projected along the surface of a smooth inclined plane with speed 10 m/s along the direction shown at $t = 0$. The point of projection is origin, z-axis is along vertical. The acceleration due to gravity is 10 m/s^2 . Column I lists values of certain parameters related to motion of ball and Column II lists different time instants. Match appropriately.

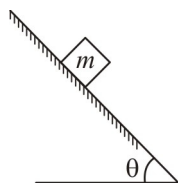
Column I

- (A) Distance from x-axis is 2.25 m
 (B) Speed is minimum
 (C) Velocity makes angle 37° with x-axis

Column II

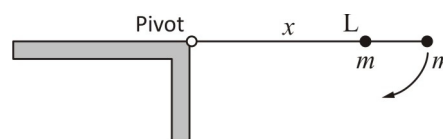
- (p) 0.5s
 (q) 1.0s
 (r) 1.5s
 (s) 2.0s

19. A block placed on a rough inclined plane. Angle of inclination θ of the plane as shown is varied starting from zero. The coefficient of static friction and kinetic friction between the block and the plane is μ_s and μ_k respectively ($\mu_s > \mu_k$). Column II shows the graphs which necessarily contains θ taken on x-axis. Column I represents the quantities taken on y-axis of column I. Match the quantities of column I with graphs of column II.

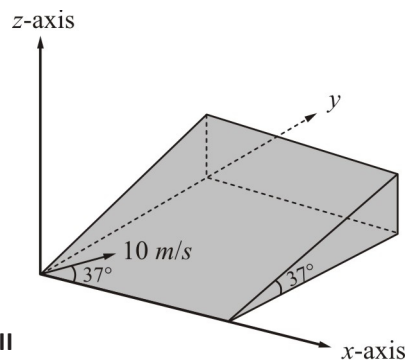


- 17.

A massless stick of length L is pivoted at one end and has a mass m attached to its other end. It is held in a horizontal position. Where should a second mass m be attached to the stick, so that the stick falls with maximum angular acceleration as possible when dropped ?



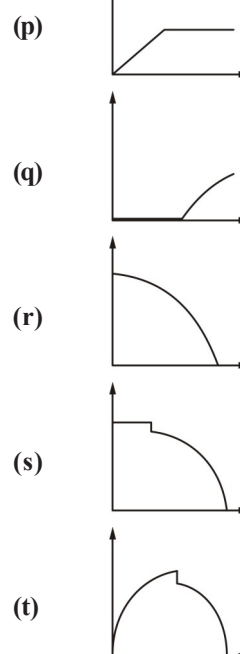
- (A) $x = \frac{L}{2}$ (B) $x = \frac{L}{2\sqrt{2}}$
 (C) $x = (\sqrt{2} - 1)L$ (D) $x = \frac{3L}{4}$



Column I

- (A) Friction force between the block and plane
- (B) Normal force between the block and the plane
- (C) Total contact force between the block and the plane

Column II



20. Consider a shell rolling purely on a horizontal fixed surface. The velocity of the centre is $v_0 \hat{i}$ and acceleration of the centre is $a_0 \hat{i}$.

Following representative points are chosen on the shell

P (on z-axis)

Q and R (on x-axis)

S (on y-axis)

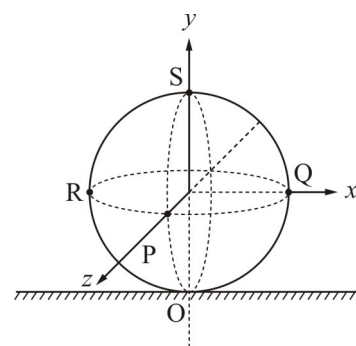
Following symbolic representation has been used :

$\vec{r}$ = position vector of a point with respect to O

$\vec{v}$ = velocity vector of point with respect ground

$\vec{\omega}$ = angular velocity of the body

$\vec{a}$ = acceleration of a point with respect to ground



Match the conditions given in column I with points in column II that obey the given condition :

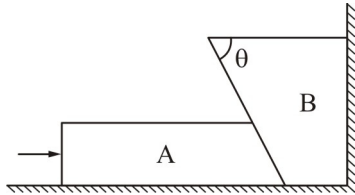
Column I

- (A) $\vec{r} \cdot \vec{a} = 0$
- (B) $\vec{r} \cdot \vec{v} = 0$
- (C) $\vec{\omega} \cdot \vec{r} = 0$

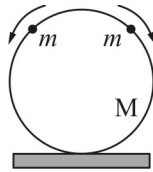
Column II

- (p) P
- (q) Q
- (r) R
- (s) S

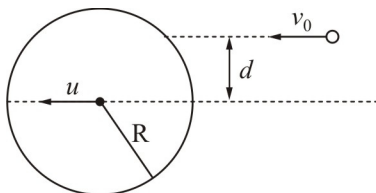
21. In the figure shown, the coefficient of friction between B and the wall is $2/3$ and the coefficient of kinetic friction between B and the wall is $1/3$. Other contacts are smooth. Find the minimum force F required to lift B, up. Now if the force applied on A is slightly increased than the calculated value of minimum force, then find the acceleration of B. Mass of A is $2m$ and the mass of B is m . Take $\tan \theta = 3/4$.



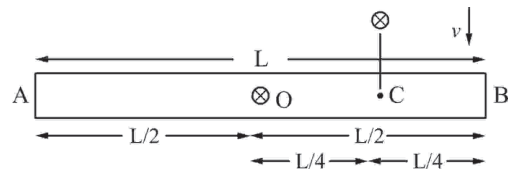
22. Two beads of mass m are positioned at the top of a frictionless hoop of mass M and radius R , which stand vertically on the ground. The beads are given tiny kicks, and they slide down the hoop, one to the right and one to the left, as shown below. What is the smallest value of m/M for which the loop will rise up off the ground, at some time during the motion is $m/2$ then find the value of m .



23. A massive disc of radius R is moved with a constant velocity u on a frictionless table. Another small disc collides with it elastically with a speed of $v_0 = 0.3 \text{ m/s}$, the velocities of the discs are parallel. The distance d shown in the figure is equal to $R/2$, friction between the disc is negligible. For which u (in m/s) will the small disc move perpendicularly to its original motion after the collision?

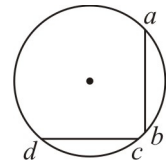


24. A homogeneous rod of length, $L = 1.8\text{m}$ and mass $M \text{ kg}$, is pivoted at the centre O , in such a way, that it can rotate freely in the vertical plane (see figure). The rod is initially in the horizontal position. A body of mass M (same as the rod) falls vertically, with speed v , on the point C , at a distance $L/4$ from one end of the rod. The mass moves towards the closer end, immediately after falling on C . Consequently, the rod rotates with constant angular velocity ω . (i) Determine the constant angular velocity (ω) in terms of v and L . (ii) if the body reaches the end (B), when the body has turned through an angle of 90° , find v .



25. The two ends of a uniform thin rod of length $\sqrt{2}R$ and of mass $2\sqrt{2} \text{ kg}$ can move without friction along a vertical circular path of radius R . The rod is released from the vertical position (ab).

Find the force exerted by an end of the path when the rod passes through horizontal position (cd).

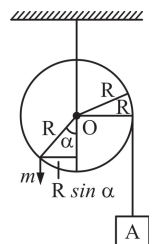


26. A uniform sphere of mass M and radius R is placed a distance $2R$ from a uniform stick of mass M and length $2R$, as shown. If these objects are released from rest, then their speeds right, before they collide due to their gravitational attraction is given by

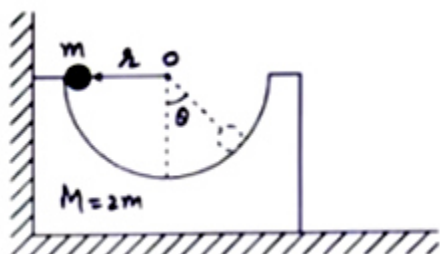
$$V = \sqrt{\frac{GM}{2R}} \ln\left(\frac{y}{2}\right) \text{ then find the value of } y.$$

27. A uniform cylindrical pulley of mass M and radius R rotates freely about a horizontal axis through O (see figure). The free end of a thread tightly wound on the pulley, carries a dead weight A .

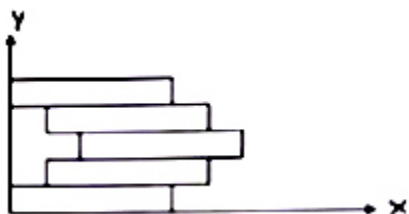
At a certain angle α , it balances a mass m , fixed on the rim of the pulley. Find the frequency of small oscillations of the arrangement.



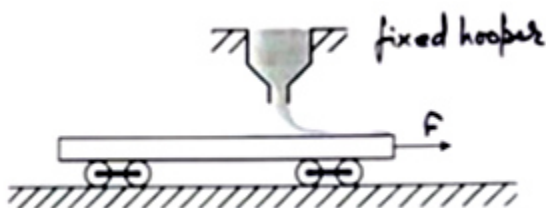
28. A ball of mass m is released from top of a cylindrical groove made in a bigger block placed on a smooth surface of mass $M = 2m$ as shown. During motion find the maximum angle with vertical to which ball can rise on the other half of groove with respect to block.



29. A bullet of mass m moving at speed v hits a ball of mass M kept at rest. A small part of ball of mass m_1 breaks from the ball and sticks to the bullet. The remaining ball is found to move at a speed v_1 in the direction of the bullet. Find the velocity of the bullet after the collision.
30. Figure shows five uniform bricks each of length l . Each brick is placed over another with an offset of length $l/10$. Find the x-coordinate of centre of mass of this system.

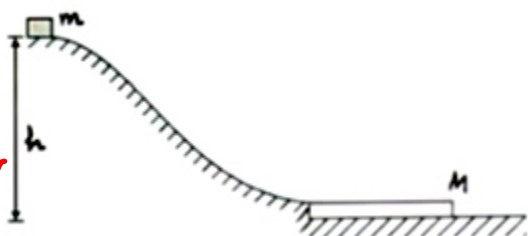


31. Figure shows a fixed hooper through which sand spills on a flat rail road car at a constant rate of r kg/sec. Car is pulled on smooth surface by a constant force F . If mass of car is m_0 find the time function of velocity and acceleration of car.



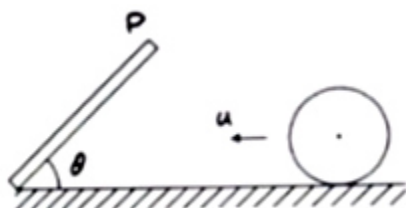
32. A sand cart of mass m_0 moves on a horizontal smooth surface due to a constant force F . At $t = 0$ cart starts from rest and through a hole in bottom sand spills at a constant rate r kg/s. Find the acceleration and velocity of cart as a function of time t .
33. A spaceship of mass m_0 moves in the absence of external forces with a constant speed v_0 . To change the direction of motion, gases are ejected from rocket in a direction normal to its motion at velocity u relative to spaceship. Find the angle by which the direction of motion of spaceship changes when its mass decreases to m .
34. A rocket starts moving from rest in space in absence of external forces by ejecting gases from a steady jet at constant velocity u relative to rocket. Find velocity of rocket when its mass is m . Given that initial mass of rocket is m_0 and mass rate of ejection of gases from jet is r .
35. A particle of mass m_1 collides elastically with a stationary particle of mass m_2 ($m_1 > m_2$). Find the maximum angle through which the striking particle may deviate as a result of collision.
36. A shell flying with velocity 500 m/s bursts into three identical fragments so that kinetic energy of the system increases 1.5 times. What maximum velocity can one of the fragments can attain?
37. A ball moving in a straight-line collides elastically with another stationary ball of same mass such that at the moment of impact the angle between the line passing through balls centre is 45° with the direction of initial motion of striking ball. If balls are smooth, find maximum fraction of kinetic energy that transform to potential energy in process of collision.
38. A particle of mass m collides elastically with another particle of mass $2m$ at rest with some impact parameter such that after collision the two particles diverge symmetrically. Find the angle of divergence.
39. A particle of mass m_1 experienced a perfectly elastic collision with a stationary particle of mass m_2 . Find the fraction of kinetic energy lost by striking particle if it recoils at right angle to original direction of motion.

40. Figure shows a small body of mass m released from a height h on a smooth hill and at bottom of hill body reaches on a plank of mass M placed on smooth ground. Due to friction between body and plank body slows down and after some time move as one piece with plank. Find the work done by friction forces in this process of motion.



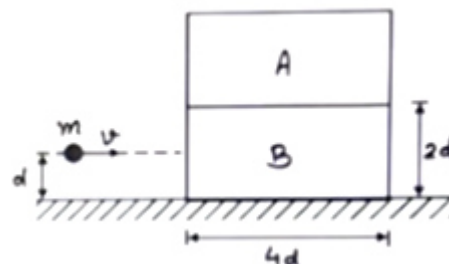
41. A steel ball of mass m falls from a height h on a horizontal surface of a table. Find the total momentum imparted by ball to table after numerous bounces, if every impact decreases the velocity of ball by n times ($n > 1$).

42. A sphere of mass m slides with velocity u on a smooth horizontal surface on which a rigid plank P is fixed at an angle θ as shown. If collision of sphere with plank is elastic find the impulse by plank and by floor on sphere during impact.



43. In gravity free space a rocket moves with constant acceleration a due to ejection of gases and external forces are absent on it. Given that gases escape at a constant speed u relative to rocket. Find the time function according to which the mass of rocket varies with time. Initial mass of rocket was m_0 .

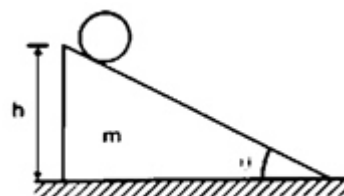
44. A block A of mass $2m$ is placed on block B of mass $4m$ which in turn is placed on a fixed table. Both blocks have same length $4d$. The coefficient of friction between table and block B is μ and there is no friction between blocks A . A small ball of mass m moving at speed v elastically collides normally to block B as shown at a height d above the table. Find minimum speed v for which block A will topple over block B .



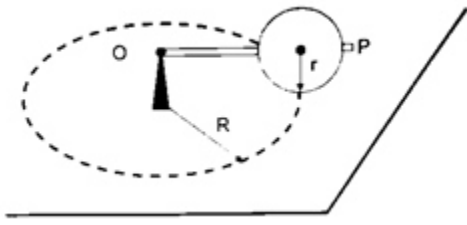
45. Figure shows two planks inclined to horizontal rough surface and connected by a string. Find the minimum angle θ for which this system remain in equilibrium if friction coefficient is $\mu = 0.5$.



46. A uniform solid sphere of mass m and radius r rolling without slipping down on an inclined wedge of same mass as shown. Find velocity of wedge when sphere reaches the bottom of wedges.

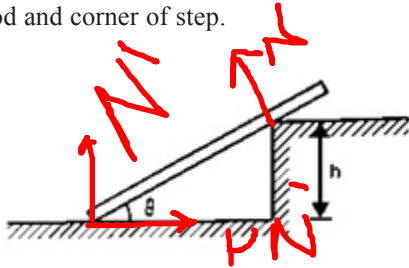


47. A sphere of mass m and radius r rolls without sliding over a horizontal plane, rotating about a horizontal axis OP . During motion the centre of sphere moves at a velocity v along a circle of radius R . Find the kinetic energy of the sphere.

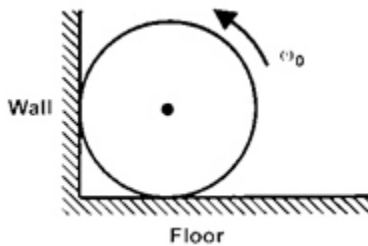


48. A rectangular plate of length l and breadth b is of mass m . Find its moment of inertia about an axis along its diagonal.

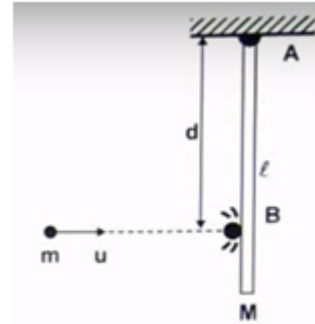
49. A uniform rod of length L rests against a step corner of height h as shown. Find the friction coefficient between ground and lower end of rod if the minimum angle the rod can rest on ground is θ . There is no friction between rod and corner of step.



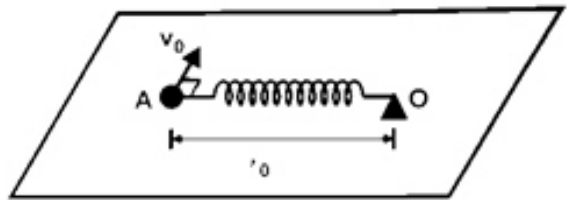
50. A uniform cylinder of radius R is spun about its axis at angular speed ω_0 and then placed into a corner. The coefficient of friction between cylinder, wall and floor is μ . Find the total number of turns cylinder will make before it stops.



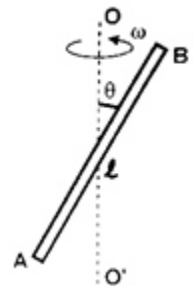
51. A uniform rod AB of mass M and length l is hinged at point A and it is hanging freely to rotate in a vertical plane. A ball of mass m moving horizontally hits it at a distance d from hinge as shown. Find d for which during collision hinge will not exert any impulse on rod.



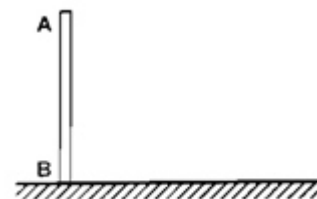
52. One side of a spring of natural length l is fixed at point O and another end of spring is attached to a small disc A of mass m . From its natural length disc is projected at speed v_0 in direction normal to the length of spring as shown. Find the force constant of spring if maximum elongation of spring in motion of disc is x .



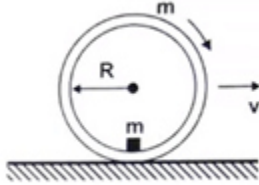
53. Figure shows a thin uniform rod AB of mass m and length l rotating about the vertical axis at angular velocity ω . Find the torque of centrifugal force acting on rod about its mid point c in the frame of reference rotating at angular speed ω attached to axis.



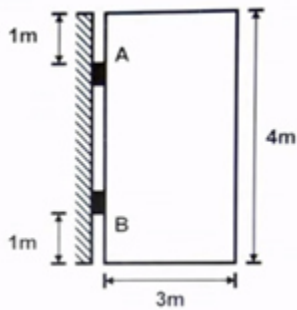
54. A thin uniform rod AB is vertically placed on a rough ground as shown. If its upper end is slightly tapped so that it starts falling and due to friction its lower end does not slip. Find the normal reaction and friction at B when rod is at an angle θ to vertical during fall.



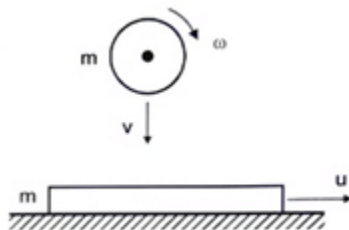
55. A small mass m is attached to a ring of same mass and radius R . The ring rolls without sliding over a horizontal plane. When the attached mass is at lowest position ring centre is moving at speed v . At what maximum value of v the ring moves without bouncing



56. A door of 100 kg mass which is supported by two hinges A and B on a wall is of size $4\text{m} \times 3\text{m}$ as shown. If the force by hinges on door are equal, find these forces exerted by hinges.



57. Figure shows a plank of mass m moving at speed u on a smooth surface. A spinning ball (sphere) of mass m and radius r normally strikes it at speed v . The friction coefficient between ball and upper surface of plank is $\mu = 0.5$ and coefficient of restitution is $e = 0.5$. Find velocity and angular velocity of sphere just after collision.



58. On a smooth rod PQ a sleeve A of mass M can smoothly slide and a thin uniform rod of mass m and length l is hinged at one end which can rotate in the plane of figure on sleeve and kept in vertical position as shown in figure. If system is released from rest, find the velocity of sleeve when rod rotates by 180° angle.

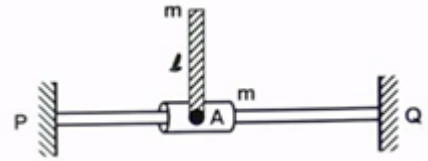
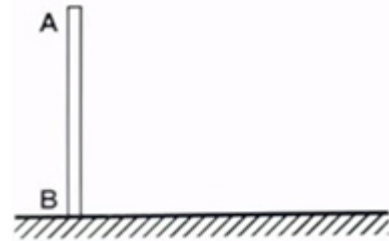
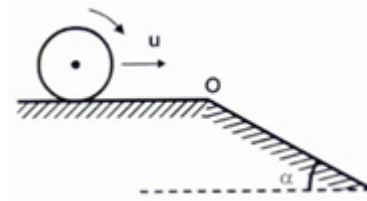


Figure shows a uniform rod AB of mass m and length l placed vertically on a rough ground. Its upper end is tapped slightly so that it starts falling and there is sufficient friction present at its lower end so that it does not slip. Find the acceleration of rod's centre of mass when it is at angle θ to vertical during motion.



60. A cylinder of radius R is in pure rolling at speed u on a horizontal plane which is passing into an inclined plane of inclination angle α as shown in figure. Find maximum value of u for which cylinder moves over to inclined plane without any jump at point O.



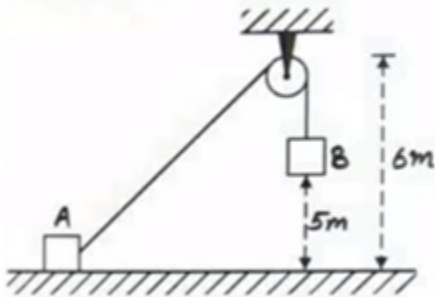
61. Figure shows a belt of mass m on two rollers of a tractor. If moves at speed v find the kinetic energy of the belt.



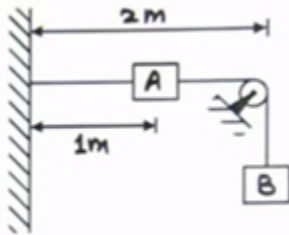
62. Figure shows a fixed hemisphere of radius R and person pulls a small body of mass m slowly by a string along the rough surface of hemisphere from ground to top of hemisphere. If friction coefficient is μ find the work done by man in the process.



63. A block A of mass m is held at rest on a smooth horizontal floor. A smooth light pulley is fixed at a height 6 m above the ground over which a string of length 16 m connected to A as shown in figure. On the other end of string a block B of same mass m connected and hanging at a height of 5 m above floor. If system is released, find the speed at which block B hits the floor.

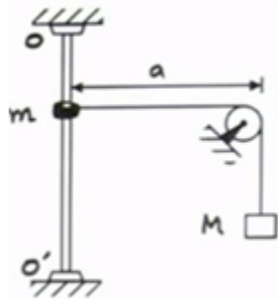


64. Figure shows a system held at rest. The masses of block A and B are 2 kg and 0.5 kg . If the system is released, find with what speed block A will hit the wall.



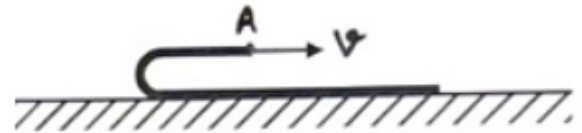
$\mu = 0$.

65. A ring of mass m which can slide on a smooth vertical rod OO' is attached to a block of mass M ($M > m$) as shown in figure. If the system is released from rest from the state shown, find the distance fallen by ring before it comes to rest for the first time.

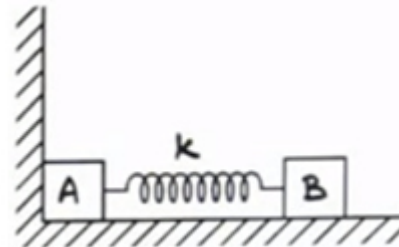


66. Two blocks of masses m , and ma connected by a non-deformed light spring rest on a rough ground as shown in figure. If friction coefficient on ground is μ find the minimum constant force F to be applied on 012 so that start sliding.

67. Figure shows a rope resting on rough ground. One end A of rope is pulled by an external force such that it moves at a uniform speed v . If mass per unit length of rope is $\lambda\text{ kg/m}$, find the power associated with the external force pulling the rope. Assume there is no slipping of rope on ground.



68. Figure shows two blocks A and B each of mass 2 kg are connected to ends of an ideal spring having force constant $k = 1000\text{ N/m}$. System of these blocks is placed on a rough floor having friction coefficient μ . Block B is pressed towards left and released. Find initial compression of spring so that block A breaks off from the wall during motion.

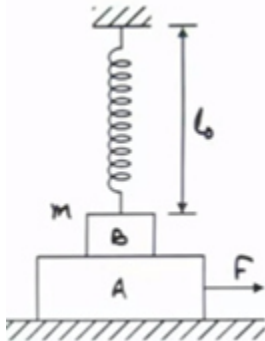


69. A chain of mass m and length l rests on a rough table with one end is hanging over the edge as shown. The friction on table is such that chain starts sliding all by itself when $x = l/3$. Find that total work done by friction on chain till chain completely slides off.

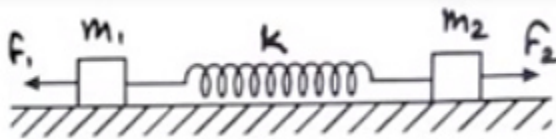


70. A body of mass m is projected on a rough horizontal surface with velocity v at $x = 0$. The friction coefficient on surface is varying with distance at $\mu = bx$. Find the maximum instantaneous power of friction force during motion of body and instantaneous power for friction force as a function of x .

71. Figure shows a system of two bodies A and B of mass m each. Ground is smooth whereas the friction coefficient between A and B is μ . Initially spring is in its natural length l_0 . A force F slowly drags the bodies to a position where B starts sliding over A and it happens when spring makes an angle θ with its initial position. Find work done by external force from start to this position when B starts sliding.



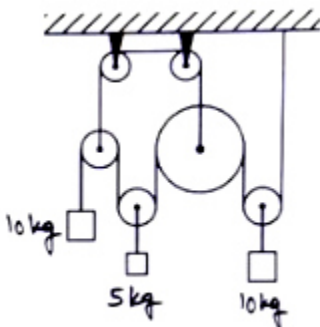
72. Figure shows two blocks of masses m_1 and m_2 attached to spring and placed on smooth ground. If two external forces act on the blocks as shown at $t = 0$ find the maximum elongation of spring during motion of blocks.



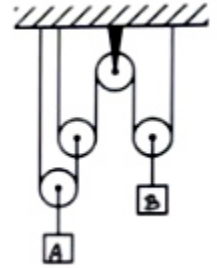
73. In figure shown the bead of mass m is free to slide on the string with some friction. If rod mass $M > m$ and system is released from rest and at $t = 0$ bead is located in front of one end of rod as shown. Find the friction force between string and bead if after t seconds bead reaches the opposite end of rod.



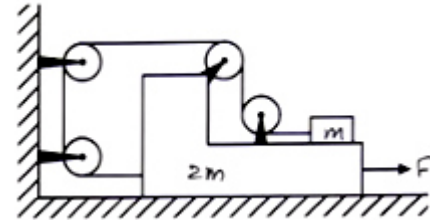
74. In system shown find acceleration of 5 kg mass.



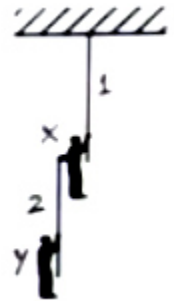
75. In figure shown if $m_A = 20$ kg and $m_B = 80$ kg find the acceleration of these blocks if system is set free to move.



76. Figure shows a system in which all surfaces are smooth. Find the acceleration of mass m .

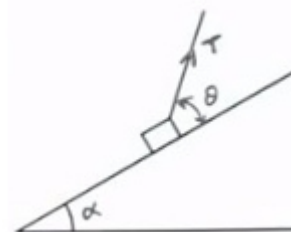


77. A boy X (mass = 20 kg) is climbing up on a strong string 1. Attached to a hook on bag of boy another string 2 is hanging which is held by another boy Y (mass = 10 kg). The breaking strength of string 2 is 200 N. find the maximum force which boy X can exert on string 1 to carry boy Y with him.

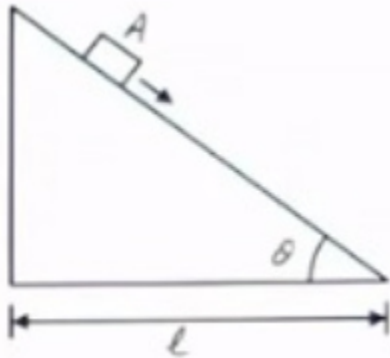


78. A body is launched up an inclined plane with inclination to horizontal. It is observed that the time of ascent is half of the time of descent along length of incline. Find the coefficient of friction between body and incline.

79. Figure shows a body of mass m pulled by a string up the incline. Friction coefficient between body and incline is μ . To slide the body find the angle at which tension in thread will be minimum.

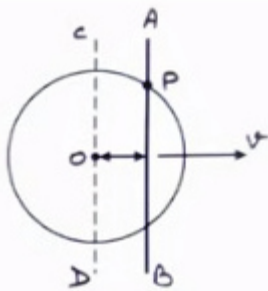


80. Figure shows a fixed wedge block of base length l and inclination θ to horizontal. A small body A is released from top of inclined surface on which friction coefficient is k . Find the value of k for which the time of sliding on inclined plane for body A will be minimum.

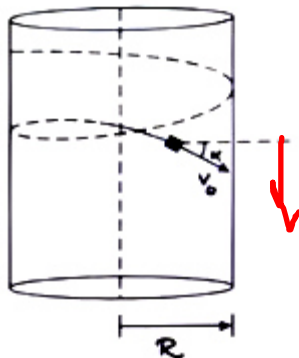


81. A particle moves at uniform speed on a parabolic trajectory $y = ax^2$ at uniform speed v . Find the acceleration of particle when it passes point $x = 0$ and point $(1, a)$.

82. A rigid Rod Moving on a Circle: A rod AB is moving on a fixed circle of radius R with a constant velocity v as shown in figure. P is the point of intersection of rod and the circle. At an instant rod is at a distance $x = \frac{3R}{5}$ from centre of circle. The velocity of rod is normal to its length and rod always remain parallel to the diameter CD. Find the speed of point P at this instant.



83. Figure shows a particle of mass m moves along inside smooth surface of a vertical cylinder of radius R . Find the contact force between particle and cylinder when its speed is v_0 and forms an angle α with horizontal.

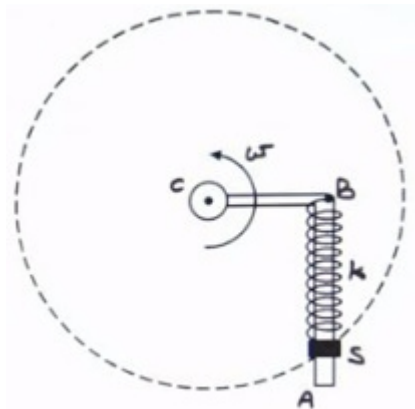


84. A car moves at uniform speed along a curve $r = b e^{\alpha \theta}$ where b and α are positive constants. The friction coefficient between road and tyres is μ . Find the maximum possible speed of car for its motion along curve without skidding.

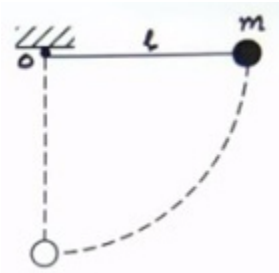
85. A particle is projected from ground with initial speed u and projection angle θ from horizontal. Find the radius of curvature of trajectory of particle when during flight particle's velocity makes an angle ϕ with horizontal.

86. A rider on a bike takes a turn on a circular road where the friction coefficient depends on the radial distance from centre r as $\mu = \mu_0 \left(1 - \frac{r}{b}\right)$ where μ_0 and b are constants. Find the radius of circle in which rider can take a safe turn with maximum velocity. Also find this maximum velocity.

87. A bent rod ABC is pivoted freely at point C about which it rotates at uniform angular speed ω in horizontal plane as shown in figure. On arm AB a spring of natural length l and force constant k is attached at point B and on its other end a movable sleeve S is attached which can smoothly slide on rod. Find the elongation of spring when sleeve is in equilibrium on rod.

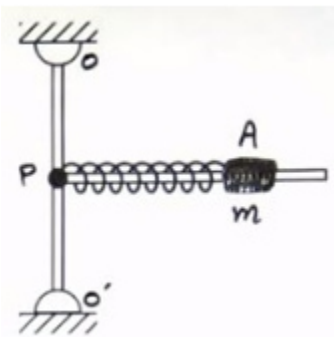


88. A bob of mass m is attached to a string of length l is taken to the position as shown where string is horizontal and released. Find the tension in string where the vertical component of bob's velocity is maximum during motion.

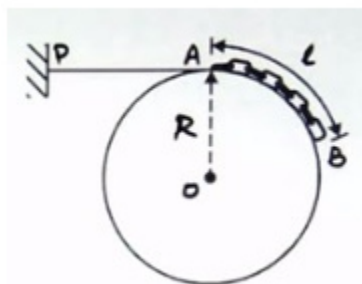


89. Find the force (magnitude and direction) acting on a particle of mass m moving in xy plane according to relation $x = b \sin \omega t$ and $y = b \cos \omega t$.

90. Figure shows a horizontal rod mounted on axis OO' and a sleeve A of mass m attached to point P by a spring of natural length l_0 and force constant k . Find the work required in rotating the rod and attaining an angular speed ω .



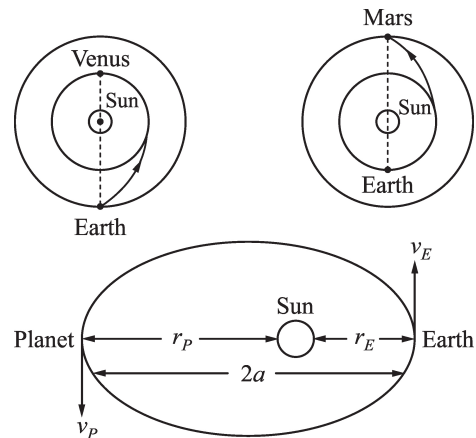
91. A chain AB of length l is placed on a smooth spherical surface of radius R with one of its ends attached to a fix point P by a thread as shown in figure. Find the acceleration of chain when thread is burnt.



92. A pendulum bob is suspended by a string of length l . From its lowest position it is thrown with a horizontal velocity v_0 such that some where string slacks and particle passes through the point of suspension following the parabolic trajectory. Find v_0 .

93. A uniform hollow sphere has internal radius a and external radius b . Taking the potential at infinity to be zero, show that the ratio of the gravitational potential at a point on the outer surface to that on the inner surface is :

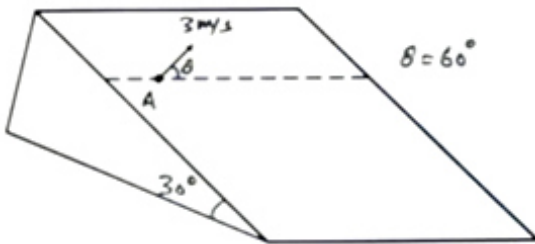
94. Several planets of the solar system have been explored using unmanned vehicle or probes. First the vehicle must acquire the earth's escape velocity, then it must be placed in a solar orbit under the sun's gravitational attraction which carries the vehical to a rendezvous with the planet. The path of the vehicle has the earth at launch and the planet at arrival on a line that passes through the sun. These paths are called Hohmann orbits and are elliptical with the sun at one focus. What is the velocity v_E of the probe from the earth, relative to sun and the velocity the probe, v_P , relative to the sun as it approaches the planet ?



95. If the planet was suddenly stopped in its orbit supposed to be circular, show that it would fall onto the sun in a time $\left(\frac{1}{4\sqrt{2}}\right)$ times its time period.
96. A satellite of mass $2m$ is in a circular orbital of radius $2R$ around the earth. By mistake, another satellite of mass m put in the same orbit and having opposite sense of rotation. The collision between the two satellite is perfectly inelastic. Will the combined mass hit the earth's surface? If yes, at what angular to the horizontal? If no, what is the minimum distance from the surface of the earth.

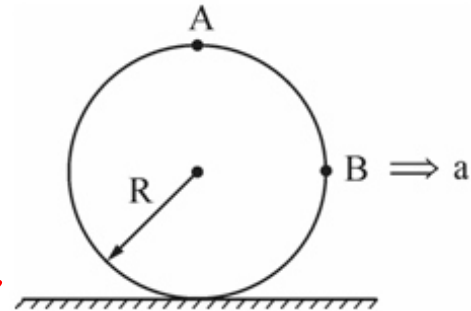
97. A particle starts moving rectilinearly at $t = 0$ such that its velocity v changes with time t according to relation $v = t^2 - t$. Find the time interval for which particle retards.
98. A railway track runs parallel to a road until a turn brings the road to a railway crossing. A cyclist rides along the road everyday at a constant speed of 20 km/hr . He normally meets the train that travels in same direction at the crossing. One day he was late by 25 min and met the train 10 km before the railway crossing. Find the speed of the train.
99. A stone is dropped from the top of a cliff of height h . After n seconds a stone is projected down from the same cliff with speed u . Show that the two stones will reach the ground together if $8h(u - gn)^2 = gn^2(2u - gn)^2$. What should be the limiting value of n in this case.
100. A train stops at two stations d distance apart and takes time t seconds from one station to another. If it first accelerates at acceleration α then decelerates at retardation β , show that $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{t^2}{2d}$.
101. A jungle safari Jeep has its front windscreen tiltable. When hailstones fall vertically at speed $5\sqrt{3} \text{ m/s}$ after bounding off from the windscreen it is observed from jeep that hailstones rebound vertically off the windscreen is tilted at angle 60° with vertical find speed of jeep. Consider elastic collision.
102. A particle moving in 2D motion has its position vector varying with time given as $\vec{r} = at^2\hat{i} + bt^4\hat{j}$. Find the equation of trajectory of particle and the angle between its velocity and acceleration vector as a function of time.
103. A body falling freely from a given height H hits an inclined plane in its path at a height h above ground. After impact the direction of velocity of the body becomes horizontal and lands on horizontal ground. Find h/H for which the body will take maximum time in reaching ground.
104. A particle projected from origin at some angle to horizontal passing through points (a, b) and (b, a) referred to horizontal and vertical axis through points of projection. Find the horizontal range and angle of projection.
105. Between two points X and Y on a straight road consistently are departing from Y to X and X to Y at uniform speed every T minutes. A cyclist riding from X to Y at 10 km/hr . Every 18 minutes a bus overtakes him in direction of motion and crosses him every 6 minutes in opposite direction. Find the speed of and time between departure of two buses.
106. A berry is hanging on a tree at a height 13 m above ground and to hit a boy standing 50 m from the tree throws a stone at $10\sqrt{10} \text{ m/s}$. If he wants to hit the berry in minimum time find the angle of projection.
107. Figure shows two parallel and inclined planks A and B at angle α to horizontal. From a point M , midway between planks a body is projected such that it grazes plank A and hits plank B normally. Find angle of projection.
108. At $t = 0$ a bird starts flying horizontally from the top of a tower of height 1.25 m with initial acceleration 1.5 m/s^2 . At the same instant a stone is projected from point B on ground as shown. If stone hits bird during its downward motion at an angle 45° to the horizontal, find the time at which stone will hit the bird.
109. A particle moves in 2D motion in such a way that its position co-ordinates are given as $x = a \sin \omega t$ and $y = a(1 - \cos \omega t)$ where a and ω positive constants and t is time. Find the equation of trajectory of particle and the distance travelled by particle as a function of time t .
110. A large chamber is sliding on a smooth inclined plane. Inside chamber from a point P a particle is projected as shown with speed u relative to chamber at an angle α . If the large range of particle is zero as seen by an observer on ground find initial speed of chamber when particle is projected.
111. A balloon starts rising from earth surface with constant speed v_0 . Due to wind the balloon gets a horizontal velocity component $v_x = ay$ where a is a positive constant and y is its height above earth surface. Find the horizontal drift of balloon as a function of height.
112. A particle moves in a plane in XY co-ordinate system with velocity vector given as $\vec{v} = a\hat{i} + bx\hat{j}$ and at $t = 0$ it starts from $(0, 0)$, find the radius of curvature of particle's trajectory as a function of x .

113. A ball is thrown from the ground to clear a wall $3m$ high at a distance of $6m$ and falls $18m$ away from the wall. Find the angle of projection of the ball.
114. A boat going downstream in a river overcame a raft at a point A at $t = 0$. At $t = 60$ minute the boat turned back and after sometime passed the raft at a distance $6km$ from point A . Find the flow velocity of river.
115. A train of length $350m$ starts from rest at uniform acceleration $0.03m/s^2$. At $t = 30s$ its head light is switched on (event-1) and at $t = 90s$ its tail light is switched on (event-2). Find the distance between these events in reference frame fixed to ground and another frame fixed to train.
116. Figure shows a fixed wedge block with its inclined surface at 30° . From point A on the inclined surface a particle is projected at $3m/s$ in the plane of inclined at an angle 60° to the line parallel to its base. Find the speed of particle after 2 second.



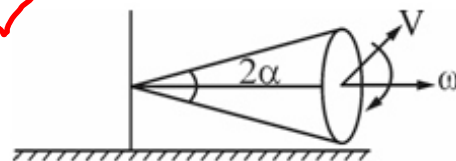
117. Figure shows two planes A and B moving in directions shows. A is moving at a speed $100\sqrt{3}m/s^{-1}$. At an instant an observer in A finds B at a distance $500m$ from it and moving in direction normal to line of motion of A and after time T from this instant plane B just miss the collision with plane A . Find T .
118. Two identical boats A and B start from a point in a river with speed v relative to water in mutually perpendicular paths, A along the river current and B perpendicular to river current. If they move equal distances from initial point and return back, find ratio of time taken by the boats A and B in round trips. Velocity of river current is $0.8v$.

119. A wheel is rolling without slipping. Find the radius of curvature of pf. A and B .



120. Two solid bodies rotate about a mutually perpendicular intersecting axes with constant angular velocities w_1 and w_2 . Find angular velocity and the angular arc of one body w.r.t. the other.

121. (i) Find angular velocity
(ii) Find angular acc. of



122. A solid body rotates with a constant angular velocity w_0 about a horizontal axis AB . At the moment $t = 0$ the AB starts rotating about the vertical axis with a constant angular acc. B_0 . Find the angular velocities and angular acc. at any time " t ".
123. A horizontal smooth rod AB rotates with a constant angular velocity " w " about a vertical axis passing through its end A . A freely sliding sleeve of mass m moves along the rod from the point A with the initial velocity " v_0 ". Find the centrifugal force acting on the sleeve at the moment when the sleeve is located at the distance r from the rotation axis.
124. A horizontal disc rotates with a constant angular velocity " w " about a vertical axis passing through its centre. A small body of mass m moves along a diameter of the disc with a velocity v_0 which is constant relative to the disc. Find the force that the disc exerts on the body at the moment when it is located at a distance " r " from the rotation axis.

Answers to Advanced Physics Assignment-1A | 2019

Mechanics

1. B 2. D 3. B 4. C 5. D 6. C 7. AC
8. BCD 9. BC 10. CD 11. AC 12. B 13. C 14. D
15. C 16. B 17. C 18. [A-p, r] [B-q] [C-s] 19. [A-t] [B-r] [C-s] [D-q]
20. [A-p] [B-p, q, r, s] [C-q, r, s] 21. 22. 3 23. 10 24. $\left(\frac{12}{7L}\right), 3.5\text{m/s}$
25. 50 26. 3 27. $\frac{1}{2\pi} \sqrt{\frac{(2mg \cos \alpha)}{[MR + 2mR(1 + \sin \alpha)]}}$ 28. $\theta = \cos^{-1}\left(\frac{1}{3}\right)$ 29. $v_2 = \frac{mv - (M - m_1)v_1}{m + m_1}$
30. $\frac{29I}{50}$ 31. $\frac{m_0 F}{(m_0 + rt)^2}$ 32. $v = \frac{F}{r} \ln\left(\frac{m_0}{m_0 - rt}\right)$ 33. $\alpha = \frac{u}{v_0} \ln\left(\frac{m_0}{m}\right)$ 34. $4 \ln\left(\frac{m_0}{m}\right)$
35. $\theta_{1\max} = \sin^{-1}\left(\frac{m_2}{m_1}\right)$ 36. 1000 m/s 37. $\frac{1}{4}$ 38. 40%
39. $f = 1 - \frac{m_2 - m_1}{m_1 + m_2} = \frac{2m_1}{m_1 + m_2}$ 40. $-\frac{mM}{m + M} gh$ 41. $m\sqrt{2gh} \left[\frac{h+1}{h-1}\right]$ 42. $J_2 = J_1 \cos \theta = 2 \mu u \cot \theta$
43. $m = \frac{36}{22} e^{-at/4}$ 44. $v = 5\sqrt{\frac{3}{2} hgd}$ 45. $\tan \theta = \frac{1}{24} = 1 \Rightarrow \theta = 45^\circ$
46. $v_2 = \sqrt{\frac{5gh}{14 \sec^2 \theta - 5}}$ 47. $\frac{7}{10} mv^2 \left(1 + \frac{2}{7} \frac{r^2}{R^2}\right)$ 48. $\frac{1}{6} \frac{M \ell^2 b^2}{(\ell^2 + b^2)}$ 49. $\frac{L \sin^2 \theta \cos \theta}{2h - L \sin \theta \cos^2 \theta}$
50. $N = \frac{\theta}{2\pi} = \frac{\omega^2 R(1 + \mu^2)}{8\pi \mu g(1 + \mu)}$ 51. $d = \frac{2\ell}{3}$ 52. $k = \frac{mv_0^2}{x^2} \left(1 - \frac{\ell_0^2}{(\ell_0 + x)^2}\right)$
53. $\frac{1}{12} m \omega^2 \ell^2 \sin \theta \cos \theta$ 54. $f = \frac{3mg}{4} \sin \theta (3 \cos \theta - 2)$ 55. $v = \sqrt{8gR}$ 56. $250\sqrt{13}N$
57. $\omega_1 = \frac{15}{4} \frac{\mu v}{r}$ 58. $v = \sqrt{\frac{3g\ell}{5}}$ 59. $\frac{3g}{4} \sqrt{5 + 3 \cos^2 \theta - 8 \cos \theta}$
60. $u = \sqrt{\frac{gR}{3}} (7 \cos \alpha - 4)$ 61. mv^2 62. $(\mu + 1)mgR$ 63. $v = \frac{40}{\sqrt{41}} \text{m/s}$
64. 3.4 m/s 65. $y = \frac{2mMa}{(M^2 - m^2)}$ 66. $F = hg \left(m_2 + \frac{m_1}{2}\right)$ 67. $P = Fv = \frac{1}{2} \lambda v^3$
68. $\frac{10 \pm 20}{1000} = 0.03m$ 69. $\omega = -\frac{1}{9} mg\ell$ 70. $\frac{1}{2} m v_0^2 \sqrt{bg}$ 71. $W = \frac{\mu mg \ell_0 (\sin \theta - 1)}{2(\sin \theta + \mu \cos \theta)}$

$$72. \ell_m = \frac{2}{k} \left(\frac{m_1 F_2 + m_2 F_1}{m_1 + m_2} \right)$$

$$73. f = \frac{2mMl}{(M-m)t^2}$$

$$74. a_2 = \frac{70}{13} \text{ m/s}^2$$

$$75. a_1 = \frac{140}{34} = \frac{70}{17} \text{ m/s}^2, a_2 = \frac{140}{17} \text{ m/s}^2$$

$$76. a = \frac{F}{3m}$$

$$77. T_1 = 300 + 300 = 600 \text{ N}$$

$$78. \mu = \frac{3}{5} \tan \alpha$$

$$79. \theta = \tan^{-1}(\mu)$$

$$80. \alpha = \frac{1}{2} \cot^{-1}(-k)$$

$$81. a_N = \frac{v^2}{r} = \frac{2av^2}{(1+4a^2)^{3/2}}$$

$$82. \frac{5v}{4}$$

$$83. N = \frac{mv_0^2 \cos^2 \alpha}{R}$$

$$84. \alpha \sqrt{\frac{\mu g}{b}}$$

$$85. r = \frac{u^2}{g} \cos^2 \theta \sec^3 \frac{\theta}{2}$$

$$86. \frac{1}{2} \sqrt{\mu_0 b g}$$

$$87. x = \frac{m\omega^2 l}{k - m\omega^2}$$

$$88. T = \sqrt{3}mg$$

$$89. \vec{F} = -m\omega^2 \vec{r}$$

$$90. \omega = E = \frac{1}{2} k \left(\frac{m\omega^2 l_0}{k - m\omega^2} \right)^2 + \frac{1}{2} m \left(l_0 + \frac{m\omega^2 l_0}{k - m\omega^2} \right) \omega^2 \quad 91. \text{Tangential acceleration of chain } a_t = \frac{F_t}{m} = \frac{gR}{l} \left[1 - \cos \left(\frac{l}{R} \right) \right]$$

$$92. v_0 = \sqrt{(2 + \sqrt{3})gl}$$

$$97. \frac{1}{2} \text{ sec} < t < 1 \text{ sec}$$

$$98. u = 75 \text{ m/s}$$

$$99. 8h(u - gn)^2 = gn^2(2u - gn)^2$$

$$100. \frac{1}{\alpha} + \frac{1}{\beta} = \frac{t}{v_m} = \frac{t^2}{5d}$$

$$101. 15 \text{ m/s}$$

$$102. \theta = \cos^{-1} \left[\frac{4a^2 t + 48b^2 t^5}{\sqrt{4a^2 t^2 + 16b^2 t^6} \sqrt{4a^2 + 144b^2 t^4}} \right]$$

$$103. \frac{h}{H} = \frac{1}{2}$$

$$104. \theta = \tan^{-1} \left[\frac{a^2 + ab + b^2}{ab} \right]$$

$$105. 9 \text{ min}$$

$$106. \theta = \tan^{-1} \left(\frac{3}{5} \right)$$

$$107. \tan^{-1} [\cot \alpha (\sqrt{2} - 1) + \alpha]$$

$$108. t = \frac{30}{10} = 3 \text{ sec}$$

$$109. a\omega t$$

$$110. 10.77 \text{ sec}$$

$$111. \frac{40}{3} \text{ m}$$

$$112. \left(\frac{a}{b} \right) \left[1 + \frac{b^2 x^2}{a^2} \right]^{3/2}$$

$$113. \theta = \tan^{-1} \left(\frac{2}{3} \right)$$

$$114. u = 3 \text{ km/hr}$$

$$115. \Delta \ell_t = 350 \text{ m}; \Delta \ell_g = 363.5 - 121.5 = 242 \text{ m}$$

$$116. 7.55 \text{ m/s}$$

$$117. T = 5 \text{ s}$$

$$118. \frac{5}{3}$$